THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A leaf is kept in the dark for 48 hours, then half of it is covered with black paper and kept in sunlight for 6 hours. Which statement is correct? 1

(a) the covered half turns blue-black
(b) the covered half remains brown with iodine
(c) both turn blue-black
(d) both remain brown

2. A ray of light falls on a plane mirror along the normal. The ray: 1

(a) travels along the mirror surface **(b)** is absorbed
(c) returns along the same path **(d)** passes through the mirror

3. If the Earth's axis were tilted by 45° instead of 23.5°, the seasons would be: 1

(a) absent **(b)** milder **(c)** more extreme **(d)** the same

4. Which of these energy changes occurs in photosynthesis? 1

(a) sound to light **(b)** chemical to light
(c) heat to light **(d)** light energy to chemical energy

5. During a solar eclipse, the Moon's shadow falls on the Earth. The region with total darkness is the: 1

(a) halo **(b)** penumbra **(c)** corona **(d)** umbra

6. A year in which the Earth makes exactly 366.25 days in an orbit would need: 1

(a) no leap years **(b)** more frequent leap years
(c) same **(d)** fewer leap years

7. A farmer notices that his legume plants in poor soil have swollen nodules in roots. This indicates: 1

(a) insects **(b)** nitrogen-fixing bacteria
(c) water logging **(d)** fungi attack

8. Why does the image in a mirror appear to be as far behind as the object is in front? 1

(a) mirror is thick
(b) virtual image position follows the law of reflection
(c) mirror is smaller
(d) light bends

9. A student claims that the Earth's shadow is formed only during lunar eclipses. The correct statement is: 1

(a) the Earth always has a shadow but the Moon enters it only during eclipses
(b) the shadow forms only then
(c) no shadow exists
(d) it is formed by the Moon

10. A variegated leaf kept in sunlight: only the green part turns blue-black. This shows: 1

(a) CO_2 is not needed **(b)** water is not needed
(c) chlorophyll is needed **(d)** light is not needed

11. If the angle of incidence of a ray on a plane mirror is increased by 15°, the angle of reflection: 1

(a) stays the same **(b)** decreases by 15° **(c)** doubles **(d)** increases by 15°

12. If the Earth rotated twice as fast, the day would be about: 1

(a) 48 hours **(b)** 12 hours **(c)** 6 hours **(d)** 24 hours

13. **Assertion (A):** Photosynthesis takes place in the presence of sunlight and chlorophyll. 1
Reason (R): Chlorophyll absorbs the energy of sunlight needed to make food.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A shadow is formed when an opaque object comes in the path of light. 1
Reason (R): Light travels in straight lines.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Day and night occur because of the rotation of the Earth. 1
Reason (R): The Earth rotates on its axis once in about 24 hours.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a plant in a closed jar with a burning candle and plant show that plants restore the air? 2

17. A ray is incident at 90° to a plane mirror. State the angle of incidence and the path of the reflected ray. 2

18. Explain why the Moon's light does not produce a clear shadow as strongly as the Sun's. 2

19. Explain why the roots of water-logged plants rot. 2

20. Why is the shadow of a tall pole longer in the evening than at noon? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Design an experiment to show that oxygen is released during photosynthesis, with controls. 3

22. A ray of light strikes a mirror with an incident angle of 55°. Find the angle between the incident ray and the reflected ray, and the angle between the mirror and the reflected ray. 3

23. The Earth rotates 360° in 24 h. Find how many degrees it rotates in 1 h and 4 min, and the time for 15° of rotation. 3

24. Why do trees in a dense forest have most of their leaves at the top? 3

OR

Describe how to use a stick and its shadow to find the time of noon. 3

25. Explain how a total lunar eclipse might last over an hour but a total solar eclipse only minutes. 3

OR

Why does an aquatic plant show more bubbles on a bright day? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Anita keeps a potted plant in a dark room for two days. Then she covers part of a leaf with black paper, keeps the plant in sunlight for a few hours, plucks the leaf, boils it in alcohol and adds a few drops of iodine solution. 5

(i) Why was the plant kept in the dark first?

(ii) Which solution is used to test for starch?

(iii) What colour will the uncovered part of the leaf turn?

(iv) What will happen to the covered part, and what does this show?

27. Design an experiment using a globe, a lamp and a stick to demonstrate the length of the shadow and the Sun's height at different latitudes. 5

OR

Describe an investigation using three potted plants (light/dark, with/without water, with/without CO_2) to study the factors of photosynthesis. 5

END OF QUESTION PAPER — Total: 50 Marks

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